25900359 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) Regular EXAMINATION October 2025

Third SEMESTER

Discipline Specific Elective (DSE) - MG3DSEECT201 - INTRODUCTION TO MECHATRONICS

(2024 ADMISSION ONWARDS)

Duration: 2 Hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Multiple Choice Questions

Answer all questions

Each question carries **1** marks

1. What is the role of resistors in R-2R ladder DAC? [U] / [CO1]

   a). Generate heat

   b). Form a voltage divider network

   c). Amplify signal

   d). Provide clock pulses

2. Which of the following real-world applications is best described as a mechatronic system? [U] / [CO1]

   a). Pendulum clock

   b). Electronic fuel injection system in cars

   c). Manual can opener

   d). Water tap

3. Define the primary goal of mechatronics in manufacturing? [K] / [CO1]

   a). Increase manual labor

   b). Reduce automation

   c). Enhance productivity and precision

   d). Use only mechanical tools

4. Thermocouples work by: [U] / [CO1]

a). Emitting light b). Using magnetism

c). Temperature difference between metals d). Resistance variation

5. Sensor sensitivity is important because it: [U] / [CO1]

a). Reduces size b). Determines smallest detectable change

c). Boosts output d). Speeds up signals

6. Define the primary function of a hydraulic actuator? [K] / [CO2]

a). Generate heat b). Store fluid

c). Convert fluid power into mechanical motion d). Control speed

7. Explain what is the common use of a relay in home automation? [U] / [CO2]

a). Speed control b). Voltage measurement

c). Controlling large appliances remotely d). Generating electricity

8. Proximity sensor that works on the principle of electromagnetic induction is [A] / [CO2]

a). Capacitive b). Ultrasonic

c). Inductive d). Infrared

9. Identify which gear system is suitable,If a machine requires high-speed reduction ? [U] / [CO2]

a). Spur Gear b). Helical Gear

c). Worm Gear d). Bevel Gear

10. Name the actuator that produces straight-line motion? [K] / [CO2]

a). Servo motor b). DC motor

c). Rotary actuator d). Linear actuator

11. The main advantage of a closed-loop system is: [U] / [CO3]

a). Simplicity

b). Higher accuracy and automatic correction

c). Lower cost

d). No need for sensors

12. SPI stands for: [U] / [CO3]

a). Serial Peripheral Interface

b). Special Port Interface

c). Simple Peripheral Input

d). Serial Port Integration

13. A system that changes when time is shifted is: [U] / [CO3]

a). Time-invariant

b). Time-variant

c). Causal

d). Linear

14. A signal that exists only at specific time intervals is [U] / [CO3]

a). Continuous signal

b). Discrete signal

c). Periodic signal

d). Random signal

15. ZigBee is mainly used for: [U] / [CO3]

a). Short-range, low-power communication

b). High-speed internet

c). Satellite signals

d). Long-distance telephony

16. Why is Python often preferred in rapid prototyping? [U] / [CO4]

a). Easy syntax and rich libraries

b). Faster than Embedded C

c). Requires no hardware

d). Uses less memory

17. Which is NOT an actuator? [K] / [CO4]

a). Servo motor b). Stepper motor c). Hydraulic cylinder d). Potentiometer

18. Why are collaborative robots (cobots) important in modern industries? [U] / [CO4]

a). They replace humans completely

b). They work safely alongside humans

c). They are cheaper than all robots

d). They require no programming

19. The Arduino function to read analog input is [K] / [CO4]

a). digitalRead() b). readInput() c). analogRead() d). getValue()

20. Which control system ensures better robotic arm accuracy? [U] / [CO4]

a). ON/OFF b). Manual c). PID d). None

[1x20 = 20]

Part B

Short Answer Questions

Answer any **6** questions

Each question carries **5** marks

21. Explain why is a SAR ADC commonly used in mechatronics? [U] / [CO1]

22. Describe the importance of controllers in a mechatronic system. [U] / [CO1]

23. Identify the medium used in pneumatic systems and explain? [U] / [CO2]

24. Show how would you use a potentiometer to measure angular position? [A] / [CO2]

25. What is the role of an anti-aliasing filter? [U] / [CO3]

26. Define attenuation in signal processing. [U] / [CO3]

27. Explain how the proportional term affects system performance. [U] / [CO4]

28. Explain why Python is often used in embedded system prototyping. [U] / [CO4]

[5x6 = 30]

Part C

Essay Questions

Answer any **2** questions

Each question carries **10** marks

29. Explain Mechatronics with suitable application. [U] / [CO1]

30. Use an appropriate actuator (DC, stepper, or servo motor) for a robotic arm and justify your choice. [An] / [CO2]

31. What are the typical applications of FIR filters in signal processing? [U] / [CO3]

32. List and explain different types of proximity sensors used in automation. [K] / [CO4]

[10x2 = 20]